

Bailey E. Quinn

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<https://darkghastful.github.io/Portfolio>

PUBLICATIONS

- [1] B. E. Quinn, J. A. R. Rodríguez, E. K. Sam, et al. The nasal microbiome in early infancy is primarily shaped by the maternal nasal microbiome. *Journal of Allergy and Clinical Immunology*, Volume 156, Issue 3, 668 - 678.

EXPERIENCE

THERMO FISHER (CONTRACT WITH ASTON CARTER) | DATA SCIENTIST II

Oct 2025 - Current | St. Louis, MO

- Built and maintain a cross-platform automation workflow leveraging Microsoft Fabric, Python, SQL, and Smartsheet that generates customizable training reports.
- Designing and implementing SQL server architecture to centralize distributed datasets and provide transparency on data processing through documentation.
- Assisting in the creation of automations and dashboards to streamline existing processes.

WASHINGTON UNIVERSITY IN ST. LOUIS MEDICAL SCHOOL DIVISION OF INFECTIOUS DISEASE | BIOINFORMATICIAN

Sept 2024 - Aug 2025 | St. Louis, MO

- Designed an R package to map DNA mutations to mutated protein sequences and premature truncation using live API queries.
- Mentored graduate students and research staff in bioinformatics workflows, providing guidance on experimental design, data analysis strategies, and tool development.
- Built reusable ETL pipelines in R/Python utilizing REST APIs and headless Chrome to ingest, clean, and analyze 0.5–2 TB datasets; parameterized for reproducibility.
- Facilitated large-scale data transfers (3–9 TB) with verification.

WASHINGTON UNIVERSITY IN ST. LOUIS MEDICAL SCHOOL DIVISION OF ALLERGY AND IMMUNOLOGY | BIOINFORMATICIAN

Aug 2022 - Sept 2024 | St. Louis, MO

- Conducted analytics for a three-hospital cohort study; built ETL pipelines in R, executed PERMANOVA/ordination/clustering, and was first author on a the corresponding publication.
- Administered lab databases and object oriented sequence stores for data validation, summarization, and analysis
- Collaborated with both clinicians and researchers including the development of a conditional diffusion model assessing sample similarity.

ORGANIZATIONS

SOUTH BROADWAY ART PROJECT
MIZZOU STEEL BRIDGE TEAM
COLLEGE MENTOR FOR KIDS

SKILLS

LANGUAGES

Python • PySpark • SQL • R • Java

ANALYTICS AND ML

ETL/ELT • large scale data management/architecture • data modeling • package development • conditional diffusion modeling • experimental design • REST API integration

PLATFORMS

Spark • MYSQL/Oracle • Fabric • AWS • Docker • Git • Power BI

EDUCATION

UNIVERSITY OF MISSOURI

BACHELOR OF SCIENCE IN
BIOCHEMISTRY
2022 | Columbia, MO

PRESENTATIONS

DOCKER/18S WORKSHOP

Nov 2023 | St. Louis, MO
Created and ran an interactive workshop designed to teach the processing and storage of sequencing data, an introduction to statistical analytics, and how to use docker images on a remote server.

REGENERATIVE MEDICINE WIP

Apr 2023 | St. Louis, MO
Presented my research detailing the transmission of bacteria from the maternal nasal microbiome to the infant nasal microbiome.

ACHIEVEMENTS

BLACK AND GOLD/ALUMNI SCHOLAR

Aug 2018 - May 2022 | Columbia, MO

HONORS DISCOVERY FELLOWSHIP

Aug 2018 - May 2019 | Columbia, MO